

Geographic exposure and planting responses

$$D_{it} = \sum_r w_{ir} S_{rt}$$

Refinery policy exposure $\times$ predetermined refinery-crusher weights

$$y_{ibt} = \alpha_{ib} + \lambda_t + \sum_k \beta_k D_{it} \mathbf{1}\{b = k\} + \varepsilon_{ibt}$$

Outcome: soybean share in exclusive ring b around hub i
Hub $\times$ ring fixed effects; year fixed effects; effects may differ by ring

Planned specification: identification requires credible shock timing and exposure; outer rings are not untreated controls.